

Operating Systems and Networking

Operating Systems

IDENTIFICATION

CODE : IF-3-SYS
ECTS : 2.0

HOURS

Lectures : 9.0 h
Seminars : 16.0 h
Laboratory : 0.0 h
Project : 0.0 h
Teacher-student
contact : 25.0 h
Personal work : 25.0 h
Total : 50.0 h

ASSESSMENT METHOD

Written exam (1h30) + continuous
assessment

TEACHING AIDS

Lecture notes for the course, work
sheets for classwork and labs [in
French]

TEACHING LANGUAGE

French

CONTACT

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AIMS

The aim of this course is to get a basic understanding of the fundamental concepts and issues in the topic of operating systems: management and sharing of hardware resources, program isolation, and user interaction. Without going into implementation details, we will give an overview of the various services offered by a typical OS, and learn how to use them appropriately when writing a program.

Expected skills:

- navigate in a directory tree
- write a multithreaded program with synchronization
- leverage OS services when developing an application

CONTENT

- purpose and structure of an operating system. concept of "shell"
- concept of process: execution time sharing (thread) and address space isolation
- memory management: contiguous allocation (malloc/free), input-output (mmap)
- file systems: directory structure (VFS), volume (mounting), inodes
- synchronization: interleaving, race conditions, critical section, lock (mutex)

BIBLIOGRAPHY

- [1] Operating System Concepts Essentials, A. Silberschatz P. Galvin and G. Gagne, Wiley.
- [2] The C Language. Brian W. Kernighan, Dennis M. Ritchie.
- [3] Computer Organization and Design. David Patterson, John L. Hennessy.

PRE-REQUISITE

- mathematics: convert between decimal and binary , use the hexadecimal notation
- architecture: registers vs ALU, CPU vs bus vs memory, the Von Neumann cycle, address space vs memory size.
- programming: control flow and variables, functions, loops, simple data structures such as arrays and linked lists.
- C language: write, run and debug simple programs on linux. design data structures including pointer fields, strings.